

Age and Gender Detection Using OpenCV and Pretrained Caffe Models

Internship Project Report

Project Overview

This project implements a real-time Age and Gender Detection system using OpenCV and pretrained Caffe models. The application detects faces from images or webcam input and predicts the age group and gender of the detected person. The project was completed as part of an internship training program and focuses on implementation and practical understanding rather than developing a custom model from scratch.

Dataset

The dataset used in this project was collected from Kaggle. Only a filtered subset of approximately 100–200 images was used. Images were filtered, cropped, resized, and organized for testing. No manual image enhancement or artificial modification was performed.

Technologies Used

Python, OpenCV, NumPy, Caffe Models, Visual Studio Code, and Webcam-based testing.

Methodology

Face detection was performed using OpenCV DNN. Detected faces were cropped, resized, normalized, and passed to pretrained age and gender models. Predictions were displayed in real time.

Implementation

The application loads pretrained Caffe models, processes webcam frames, detects faces, predicts age and gender, and displays the results continuously.

Results

The system successfully detected faces and predicted age groups and gender in real time. Performance was better under good lighting and front-facing conditions.

Challenges Faced

Limited dataset size, lighting variations, side-face detection issues, and understanding OpenCV DNN workflows.

Limitations

The project uses pretrained models and a small dataset. Results are intended for learning and demonstration purposes rather than production use.

Learning Outcomes

Image preprocessing, face detection, pretrained model deployment, real-time computer vision, and GitHub project documentation.

Conclusion

This project demonstrates a practical implementation of age and gender detection using OpenCV and pretrained Caffe models and provided valuable hands-on experience during the internship.